1.11 Quadratic Modelling

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The new A Level has a particular emphasis of the application of theory to real-life situations. A mathematical model is the maths used to model such a situation, possibly with some simplifying assumptions.

Example (from textbook): A spear is thrown over level ground from the top of a tower.

The height, in metres, of the spear above the ground after t seconds is modelled by the function:
$h(t) = 12.25 + 14.7t - 4.9t^2, t \geq 0$

a) Interpret the meaning of the constant term 12.25 in the model.

b) After how many seconds does the spear hit the ground?

c) Write $h(t)$ in the form $A - B(t - C)^2$, where A, B and C are constants to be found.

d) Using your answer to part c or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached?

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2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph.

$$y = -0.01x^2 + 0.975x + 16, \ x > 0.$$

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg.
b Rewrite y in the form $A - B(x - C)^2$, where A, B, C are constants to be found.
c Using your answer to part b, find the speed at which the car has the greatest fuel efficiency.
d Use the model to calculate the fuel consumption of a car travelling at 120 mph.
Comment on the validity of using this model for very high speeds.

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3 A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield g, measured in tonnes per hectare.

$$g = 6 + 0.03f - 0.00006f^2.$$

a According to the model, how much grain would each hectare yield without any fertiliser? (1 mark)
b One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? (4 marks)

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4 A football stadium has 25000 seats. The football club know from past experience that they will sell only 10000 tickets if each ticket costs \pounds30. They also expect to sell 1000 more tickets every time the price goes down by \pounds1.

a The number of tickets sold t can be modelled by the linear equation $t = M - 1000p$, where \pounds$p$ is the price of each ticket and M is a constant. Find the value of M. (1 mark)

The total revenue, \pounds$r$, can be calculated by multiplying the number of tickets sold by the price of each ticket. This can be written as $r = p(M - 1000p)$.

b Rearrange r into the form $A - B(p - C)^2$, where A, B, C are constants to be found. (3 marks)
c Using your answer to part b or otherwise, work out how much the football club should charge for each ticket if they want to make the maximum amount of money. (2 marks)